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TigCLaF: A Cross-Lingual Large Language Model Framework for Sentiment-Aware Text Classification in Low-Resource Tigrigna

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Abstract

This paper introduces **TigCLaF**, a novel cross-lingual large language model framework for sentiment-aware text classification in low-resource Tigrigna. The framework integrates tokenizer extension with adaptation, continual pretraining on unlabeled Tigrigna data, and LoRA-based parameter-efficient fine-tuning to enable effective cross-lingual adaptation from high-resource languages. Leveraging recent advances in multilingual pre-trained language models and large language models, we investigate zero-shot, few-shot, and full fine-tuning strategies for sentiment detection, incorporating transformer models XLM-RoBERTa and AfriBERTa, as well as instruction-tuned LLaMA models. Our approach integrates a Tigrigna sentiment lexicon into transformer-based embeddings via feature fusion, thereby enhancing preservation of the sentiment signal during cross-lingual transfer. The proposed framework is evaluated on a newly curated dataset of 30,000 Tigrigna instances, supported by auxiliary English and Amharic sentiment corpora for transfer learning. Experimental results show that sentiment-aware feature integration improves classification accuracy and Macro-F1 up to 7% over baseline multilingual models without sentiment augmentation.

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Furthermore, parameter-efficient fine-tuning LoRA achieved competitive accuracy while reducing model size and inference latency, making it suitable for computationally constrained settings. The systematic error analysis highlights the roles of script-specific preprocessing, idiomatic expressions, and nuances of cultural sentiment in classification performance. The proposed framework demonstrates the viability of combining LLM-based cross-lingual transfer with masked sentiment-aware enhancements for practical, resource-efficient NLP applications in low-resource language contexts.

Keywords: Tigrigna, Sentiment-Aware classification, Low-Resource Languages, Cross-Lingual Representation Learning, Multilingual Adaptation.

1 Introduction

In the era of information overload, Sentiment-aware text classification involves analyzing text to determine emotional tone [1–4]. Sentiment-aware classification preserves emotional tone, yet it remains underexplored in low-resource languages such as Tigrigna [5–8]. Recent overviews of Tigrigna NLP underscore gaps in labeled data, preprocessing tools, and task-specific models, especially for sentiment classification [1]. The state-of-the-art text classification models typically rely on large, high-quality labeled datasets, a luxury not available for Tigrigna. Cross-lingual transfer learning, leveraging models trained on resource-rich languages, offers a compelling path forward. For instance, XLNet pretrained on English data achieved up to 78.9% F1-score in classifying Tigrigna sentiment with just 10K labeled examples, outperforming both BERT and mBERT baselines [8]. Broad surveys of cross-lingual sentiment analysis underscore the growing maturity of methods utilizing machine translation, bilingual lexicons, and mBERT, XLM-R multilingual models [9]. More recent work demonstrates that multilingual lexicon-enhanced pretraining enables zero-shot sentiment analysis across dozens of low-resource languages, including code-mixed contexts, with performance surpassing even large language models like GPT-3.5 [10]. Earlier classification efforts in Tigrigna utilized word embeddings and convolutional neural networks to reach around 93% accuracy on a six-class news dataset [1]. However, domain-specific sentiment classification has received much less attention. Sentiment-aware cross-lingual methods specific to Tigrigna remain unexplored, especially regarding enhancements using sentiment lexicons and tailored attention mechanisms.

Cross-lingual pretrained language models (PLMs) and large language models (LLMs) have transformed multilingual text classification. Masked-LM encoders such as XLM-R and sequence-to-sequence models like mT5 provide strong zero- and few-shot transfer across over 100 languages, including many low-resource ones. These models achieve substantial gains on cross-lingual benchmarks by scaling training data and parameters, enabling target-language generalization with minimal or no labeled supervision [11]. Large Language models introduce two complementary capabilities for low-resource settings. First, in-context learning (ICL) enables a frozen model to perform classification using a handful of exemplars or instructions embedded in the

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prompt, dramatically reducing annotation and fine-tuning requirements. Building on seminal work showing few-shot competence in GPT-3, subsequent studies analyze and improve ICL for multilingual and cross-lingual scenarios via prompt construction and alignment strategies [12]. The second parameter-efficient fine-tuning techniques, such as LoRA and QLoRA, adapt LLMs to domain/language constraints by training lightweight adapter modules or low-rank updates while keeping the backbone frozen (and often quantized). These methods reduce memory and compute, making targeted sentiment/classification adaptation feasible on commodity hardware, a practical requirement for expert systems deployed in low-infrastructure contexts [13].

LLMs further benefit from translation-augmented pipelines: high-coverage MT (such as No Language Left Behind [14], FLORES-200 [15]) enables translate-train and translate-test strategies that convert Tigrigna inputs into a high-resource language for LLM-based classification or distill them back into a compact classifier. Recent work shows robust cross-lingual sentiment and classification improvements through MT+LLM ensembles and expert-pruned multilingual systems, suggesting a practical route for settings with scarce labels [15–17]. The rapid advance of multilingual foundation models of the LLaMA family brings larger context windows, improved multilingual pretraining, and stronger instruction following, features that directly improve prompt-based and adapter-based text classification in low-resource languages. Empirical reports highlight competitive multilingual performance at scale, providing an attractive backbone for our classification pipeline [18, 19].

The State-of-the-art text classification methods typically assume abundant labeled data and mature preprocessing, conditions that do not hold for Tigrigna. Even with strong multilingual PLMs or LLMs, transfer degrades under morphological divergence, domain shift, and scarce sentiment resources (lexicons, guidelines). Systematic reviews of cross-lingual sentiment analysis consistently identify a lack of target-language sentiment resources and alignment challenges as core bottlenecks [20]. Moreover, while ICL reduces labeling demands, cross-lingual ICL remains fragile without careful example selection and source-target alignment; recent studies show retrieval-augmented and alignment-aware ICL can mitigate these issues, but are underexplored for Ge'ez-script languages [21].

Previous studies on sentiment and topic classification in under-resourced morphologically rich languages emphasized careful preprocessing pipelines, lightweight models, and transfer from high-resource languages. For languages written in Ge'ez (Tigrigna and Amharic), tokenization and normalization are non-trivial because orthography, punctuation, and spacing conventions differ from Latin-script languages (Ge'ez uses dedicated word and sentence marks); robust pipelines typically combine Unicode-aware boundary detection (such as ICU, TR29 rules) with script-specific heuristics and normalization [22]. Community guidance, such as W3C's Ethiopic layout requirements and ICU boundary analysis, underpins practical implementations in modern NLP stacks [23]. Beyond general pipelines, several studies examined normalization for African languages and showed that finite-state rewrite rules and dataset cleanup significantly affect downstream model perplexity and accuracy, an effect amplified in low-resource settings [24]. For Tigrigna specifically, public resources have begun to emerge: monolingual PLMs such as TiRoBERTa [25], TiELECTRA [26], early

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sentiment datasets (YouTube/Twitter), and broader benchmarks as TiQuAD [27] for question answering, collectively enabling reproducible classification research and fine-tuning [8].

Parallel advances in multilingual foundation models offer improved multilingual pretraining, larger context windows, and stronger instruction-following capabilities, making them attractive backbones for prompt-based and adapter-based classification in low-resource languages [18, 19]. Cross-lingual transfer leverages multilingual pretrained language models to port classifiers from data-rich source languages English to low-resource targets [28], including mBERT, XLM-R, and mT5, and benchmarking efforts like XTREME and XGLUE established standard evaluation protocols and documented persistent performance gaps for distant or low-resource languages, gaps particularly salient for sentiment classification due to lexical and cultural nuance [11, 29, 30] can bias decision boundaries [31]. Strong baselines typically adopt XLM-R or mBERT backbones with language task-specific adapters [32, 33]. In-Context Learning (ICL) and Cross-Lingual ICL for low-resource targets, carefully curated exemplars (possibly translated), and prompt calibration can yield competitive results without fine-tuning [34]. Parameter-Efficient Fine-Tuning (PEFT) LoRA [35] and QLoRA [33] combines 4-bit quantization. Variants like LoftQ [36] improve initialization and robustness. Emerging Translation-augmented prompting and Alignment-aware prompting methods automate self-translate-train with LLMs to synthesize target-language training signals, blurring lines between prompting and data augmentation [31, 37].

Recent advances in aspect-based sentiment analysis (ABSA) for morphologically rich languages like Urdu demonstrate the effectiveness of graph-based architectures and generative frameworks [38–45]. For instance, UrduAspectNet [41] fuses transformers with dual GCNs to capture syntactic dependencies, while subsequent enhancements employ biaffine attention for improved aspect-sentiment alignment [43]. However, these approaches assume sufficient labeled data in the target language and focus on single-language optimization. Cross-lingual ABSA methods [46–48] address data scarcity through language family disentanglement [47] and LLM-based zero-shot evaluation [48], yet they often overlook script-specific challenges and parameter efficiency for deployment in low-resource languages, such as Tigrigna.

Knowledge-aware sentiment lexicon feature integration remains valuable for interpretability and robustness, especially in low-resource settings with label imbalance. Studies compare lexicon vs. transformer approaches and propose hybrids that fuse lexicon cues of polarity, valence shifters, and negation scope with contextual embeddings, improving stability under domain shift. Recent competition systems for multilingual sentiment explicitly incorporate lexicon or emotion embeddings alongside XLM-R and mBERT to handle idioms and sarcasm [49–51]. Practical sentiment lexicons for Tigrigna are still nascent; however, adapter-based multilingual models enriched with small, community-built polarity lists (or lexicons projected via cross-lingual dictionaries embeddings) can serve as sentiment priors that guide fine-tuning and calibrate decision thresholds in low-shot regimes. Hybridization is further motivated by analyses showing that BERT-like models do not always reliably attend to sentiment cue

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words without supervision, especially in morphologically complex scripts, supporting explicit feature fusion [52].

Challenges in Tigrigna sentiment analysis are both linguistic and computational. The literature suggests a convergence toward LLM-centric sentiment classification pipelines that combine (i) multilingual encoders/LLMs for cross-lingual generalization, (ii) PEFT for computational efficiency, and (iii) translation and alignment-aware prompting to mitigate language mismatch. Yet, open gaps persist for Tigrigna: (1) standardized Ge'ez-script preprocessing for classification at scale, (2) systematic evaluation of cross-lingual ICL vs. PEFT under realistic data budgets, and (3) principled integration of lightweight sentiment lexicons with LLM embeddings to improve robustness and interpretability. Our work addresses these by designing a sentiment-aware, cross-lingual text classification stack centered on mPLMs and LLMs, leveraging PEFT, translation-augmented prompts, and lexicon-informed fusion, and evaluated on public Tigrigna resources and AfriSenti derivatives [13, 31, 53]. Table 1 presents a focused comparison of recent sentiment analysis studies published between 2023 and 2025 that are most relevant to our work. To ensure clarity and comparability, we prioritize methods with clear technical overlap in cross-lingual sentiment analysis, low-resource language adaptation, and efficiency-oriented modeling, while excluding foundational surveys and infrastructure resources. The table highlights key contrasts between high-resource, language-specific ABSA models developed primarily for Urdu [41, 43, 44] and more generic multilingual or zero-shot approaches [46–48], versus our focus on sentiment classification in low-resource Tigrigna with explicit Ge'ez-script adaptation. Across all entries, we maintain consistent comparison dimensions—including modeling technique, target language, data requirements, script coverage, and parameter efficiency, allowing TigCLaF to be evaluated transparently against prior work. This selective presentation fits within a single page while preserving analytical depth and clearly situating TigCLaF's contributions in relation to recent advances.

The persistent gap in scalable NLP for underrepresented languages motivates this work, which aims to construct an effective sentiment-aware text classification pipeline for Tigrigna. Our central objectives are threefold: to build a robust classification system, to intelligently integrate cross-lingual transfer learning with sentiment feature adaptation, and to rigorously evaluate performance not only on accuracy but also on deployment efficiency, a critical consideration for real-world impact. To guide this endeavor, our investigation is structured around five pivotal research questions. We begin by asking how multilingual large language models can be effectively adapted for sentiment-aware classification in a low-resource language like Tigrigna. This leads us to probe the specific impact of tokenizer Adaptation and continual pre-training on cross-lingual alignment and sentiment understanding in morphologically rich languages. Given the practical constraints of such environments, a core question is whether parameter-efficient fine-tuning methods like LoRA can achieve competitive performance while maintaining computational efficiency. We further conduct a crucial comparative analysis to understand how fine–tuning–based approaches compare with in-context learning in terms of accuracy, stability, and interpretability. Finally, to inform future strategies, we analyze the primary sources of error, both linguistic and model-level, in this adaptation process. In response to these research questions,

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Table 1 Comparison of recent (2023–2025) sentiment analysis and aspect-based sentiment analysis approaches, grouped by methodological category and contrasted with the proposed TigCLaF framework.

Work	Task / Focus	Key Technique	Target Language(s)	Data Req.	Script	Param. Eff.
Generic and Cross-Lingual Sentiment Analysis						
[46]	Short-text sentiment analysis	Hybrid deep learning with dual attention	Generic	Medium	Generic	Medium
[47]	Cross-lingual multimodal sentiment analysis	Language-family disentanglement with transfer learning	African languages	Medium	Mixed	Medium
Zero-Shot and LLM-Based Approaches						
[48]	Zero-shot multilingual ABSA	LLM prompting (no fine-tuning)	Multiple	Low	Generic	High
High-Resource Aspect-Based Sentiment Analysis (ABSA)						
[41]	Aspect-based sentiment analysis	Transformer + dual GCN	Urdu	High	Arabic-derived	Low
[43]	Aspect-based sentiment analysis	Biaffine attention + GCN	Urdu	High	Arabic-derived	Low
[44]	ABSA quadruple extraction	Transfer learning	Urdu	Medium	Arabic-derived	Medium
Generative Sentiment and Opinion Modeling						
[39]	Generative sentiment analysis	Latent category distribution + constrained decoding	Generic	Medium	Generic	Medium
[45]	Implicit opinion mining	Term-correlation prompting	Generic	Medium	Generic	Medium
Our Proposed Framework						
TigCLaF (Ours)	**Cross-lingual sentiment analysis**	**Tokenizer adaptation + continual pretraining + lexicon fusion + LoRA**	**Tigrigna**	**Low**	**Ge'ez**	**High**

Abbreviations: SA = Sentiment Analysis; ABSA = Aspect-Based Sentiment Analysis; GCN = Graph Convolutional Network; CPT = Continual Pretraining; PEFT = Parameter-Efficient Fine-Tuning.

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our primary contribution is the introduction of TigCLaF, a novel cross-lingual framework that systematically adapts multilingual models for low-resource settings. The TigCLaF pipeline extends multilingual tokenizers with high-frequency Tigrigna subwords to address Ge'ez-script representation challenges, applies continual pretraining on unlabeled Tigrigna text to improve cross-lingual alignment, and employs LoRA-based parameter-efficient fine-tuning that updates at most 6% of model parameters. Through systematic experimentation, we compare fine-tuning, in-context learning, and translation-based strategies, analyze linguistic and model-level error sources, and demonstrate that TigCLaF achieves strong sentiment classification performance while remaining lightweight and deployable. The foundational contributions of this work are summarized as follows:

- Introduces a novel, efficiency-optimized cross-lingual framework for Tigrigna, TigCLaF, that leverages cross-lingual knowledge transfer from high-resource languages via LoRA modules for scalable model adaptation, while retaining the linguistic richness of multilingual pretrained models and improving sentiment-aware classification.
- We introduce tokenizer adaptation with Tigrigna-specific subword units to reduce out-of-vocabulary rates for Ge'ez-script text, and apply continual cross-lingual pretraining that combines masked language modeling with contrastive alignment objectives. This strategy improves semantic consistency between Tigrigna and high-resource languages (English and Amharic), yielding measurable gains in sentiment understanding and domain robustness under low-resource conditions.
- We demonstrate that combining continual cross-lingual pretraining with LoRA-based parameter-efficient fine-tuning achieves state-of-the-art sentiment classification performance for Tigrigna while substantially reducing computational and memory requirements, enabling efficient deployment in low-resource and low-compute environments.
- Introduces a new large-scale sentiment corpus for Tigrigna, curated from social media and news sources, and establishes the first reproducible benchmark for multilingual sentiment analysis in low-resource languages.

The introduction of a comprehensive, sentiment-aware, LLM-centric text classification system that integrates script normalization, multiple multilingual models, and Tigrigna-specific sentiment cues into a unified, flexible pipeline. Our study provides a detailed comparative analysis of cross-lingual strategies tailored for low-resource sentiment, directly pitting in-context learning against parameter-efficient fine-tuning and translation-based methods. Finally, with a firm eye on practical application, our evaluation is rigorously deployment-minded. We leverage PEFT and quantization to conduct a thorough efficiency–accuracy trade-off analysis and report performance metrics directly relevant to CPU-class hardware typical of real-world deployments in Tigrigna-speaking regions. Collectively, these contributions advance the goal of scalable and inclusive multilingual NLP, offering a replicable blueprint for adapting LLMs to other underrepresented African languages.

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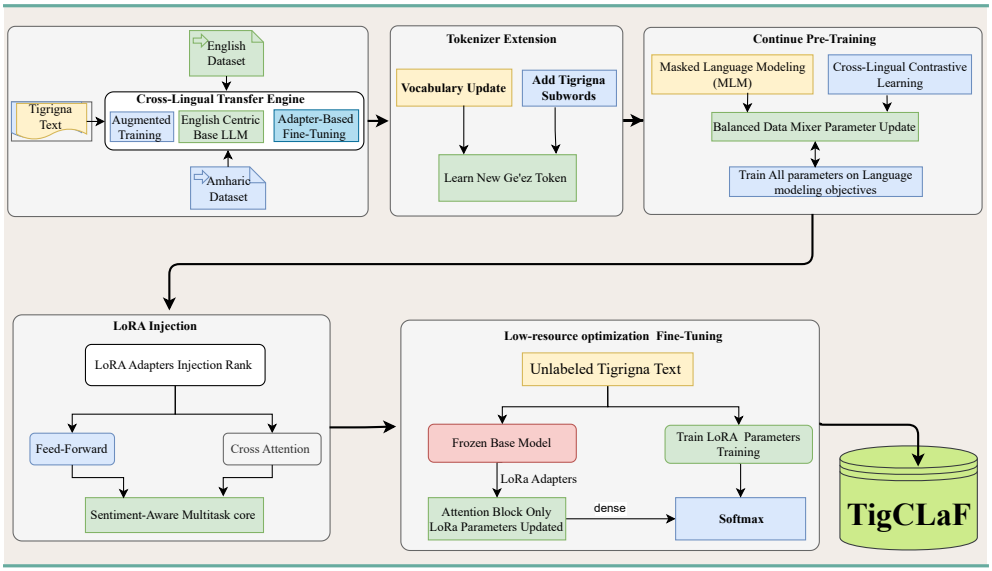

Fig. 1 The TigCLaF general system architecture. Overview of the TigCLaF proposed framework for sentiment-aware text classification in low-resource Tigrigna, integrating tokenizer extension, continual pretraining with cross-lingual alignment, and LoRA-based parameter-efficient adaptation of pretrained multilingual and large language models.

2 The TigCLaF Framework

This section presents a detailed blueprint of our sentiment-aware text classification framework for Tigrigna, emphasizing cross-lingual transfer, LLM-enabled adaptation, and practical deployment readiness.

2.1 Overview of the TigCLaF Framework

The TigCLaF framework is designed to enable effective cross-lingual adaptation of LLMs for sentiment-aware text classification in Tigrigna, a low-resource language. The proposed architecture, Figure 1, is built on top of multilingual pretrained transformers, XLM-R, mBERT, AfriBERTa, and LLaMA, which integrates three sequential phases of four key components: Tokenizer Adaptation, Continual Pretraining (CPT), and LoRA-based Fine-Tuning—organized in a modular pipeline that balances linguistic coverage, efficiency, and performance.

2.1.1 Tokenizer Extension with adaptation Layer

Tigrigna presents rich morphology and script-specific vocabulary absent in mainstream multilingual tokenizers. To bridge this gap, TigCLaF extends the base SentencePiece tokenizer by merging its subword vocabulary with the top 5,000 Tigrigna subwords, derived from a large unlabeled Tigrigna corpus. This Adaptation preserves

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orthographic fidelity and ensures minimal token fragmentation. The modified tokenizer supports code-switching and morphological decomposition, enabling smoother cross-lingual embedding alignment between Tigrigna, Amharic, and English.

2.1.2 Cross-Lingual Continual Pretraining

In this stage, the extended model undergoes domain-adaptive continual pretraining on a mixed-language corpus—consisting of 60% Tigrigna, 20% English, and 10% Amharic text—to infuse Tigrigna-specific linguistic priors into the multilingual embedding space while preserving cross-lingual generalization. Masked Language Modeling (MLM) and Cross-Lingual Contrastive Learning (CLCL) are optimized jointly as complementary objectives. This dual-objective CPT refines the multilingual representations, reducing lexical drift and enhancing cross-lingual transferability. The result is an intermediate model, M_{TigCPT} , specialized for Tigrigna text while retaining multilingual generalization.

In this stage, the extended model undergoes domain-adaptive continual pretraining on an unlabeled Tigrigna corpus to infuse Tigrigna-specific linguistic priors into the multilingual embedding space. Masked Language Modeling (MLM) and Cross-Lingual Contrastive Learning (CLCL)complementary objectives are optimized. This dual-objective CPT refines the multilingual representations, reducing lexical drift and enhancing cross-lingual transferability. The result is an intermediate model, M_{TigCPT}, specialized for Tigrigna text while retaining multilingual generalization. The total loss during Continual Pre-Training (CPT) is composed of a Masked Language Modeling loss and a cross-lingual alignment loss:

$$L_{\text{CPT}} = L_{\text{MLM}} + \lambda L_{\text{XAlign}} \tag{1}$$

where λ is a weighting hyperparameter that controls the balance between intra-lingual (L_{MLM}) and cross-lingual alignment (L_{XAlign}) objectives. Through grid search, we set $\lambda = 0.3$ to optimally align representations without degrading monolingual modeling capability.

A parallel sentence pairs are constructed via back-translation using the NLLB-200 model for cross-lingual contrastive learning. These pairs consist of machine-translated and aligned news headlines, ensuring semantic consistency across languages. The model is trained for 50,000 steps with early stopping based on validation perplexity and alignment loss stability. Our model adapts the pretrained model on unlabeled Tigrigna corpora to improve linguistic alignment. The model is further pretrained on unlabeled Tigrigna text $\mathcal{D}_{unl}$ using MLM and cross-lingual contrastive alignment losses to improve cross-lingual coherence.

2.1.3 Parameter-Efficient LoRA-Adapted Fine-Tuning Module

To ensure computational efficiency, TigCLaF employs Low-Rank Adaptation (LoRA) during supervised fine-tuning. Instead of updating all parameters, low-rank adapters of rank r = [8, 16] are inserted into the query–key–value projections of transformer attention layers and the intermediate feed-forward layers. These adapters capture task-specific variations while keeping the backbone frozen. The fine-tuning objective

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minimizes cross-entropy loss over the labeled sentiment corpus (positive, negative, neutral). For decoder-based architectures, LLaMA 2 and LLaMA 3, a prompt-style fine-tuning setup is used to align text-generation probability distributions with sentiment labels. The Parameter-Efficient Fine-Tuning enables efficient supervised adaptation using minimal trainable parameters. LoRA modules are injected into self-attention and feed-forward layers, optimizing only the low-rank matrices A and B:

$$W' = W + \Delta W = W + BA, \tag{2}$$

where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$, and r denotes the LoRA rank. This formulation enables efficient adaptation by updating only the low-rank matrices A and B while keeping the original pretrained weights W frozen, thus significantly reducing the number of trainable parameters.

2.2 The TigCLaF Architecture

The TigCLaF architecture builds upon multilingual and instruction-tuned large language models of LLaMA-2, XLM-RoBERTa, and AfriBERTa, to enable efficient cross-lingual sentiment classification for low-resource Tigrigna. These models serve as foundational encoders, providing multilingual contextual embeddings that capture both syntactic and semantic relationships across languages.

To enhance adaptability, cross-lingual adaptation layers are integrated via LoRA-based low-rank adapters that fine-tune task-specific representations without modifying the full model parameters. This design reduces computational cost while preserving multilingual generalization. Additionally, a sentiment fusion mechanism incorporates lexicon-derived polarity features into the transformer embeddings through feature concatenation and gating. This fusion enriches the model's awareness of Tigrigna-specific sentiment cues, idiomatic expressions, and contextual polarity, resulting in a sentiment-informed, cross-lingually aligned representation optimized for text classification.

The modular design leverages cross-lingual representations from high-resource languages while injecting language-specific knowledge from Tigrigna through token-level, model-level, and task-level adaptation. The backbone model serves as the foundational multilingual encoder–decoder, while the cross-lingual adaptation layers progressively specialize it for Tigrigna sentiment analysis. A lightweight adapter-enhanced architecture integrates the Tokenizer Extension with an adaptation Layer (expanding the embedding matrix with the top-k Tigrigna subword), Continual Pretraining Module(trained on unlabeled Tigrigna text using masked-language modeling (MLM) and cross-lingual contrastive objectives), LoRA Adaptation Blocks(inserted into attention and feed-forward layers for efficient fine-tuning), and a Sentiment Fusion Layer, merging lexicon embeddings with contextual representations before classification.

2.2.1 Cross-Connected Multilingual Alignment

A distinctive feature of TigCLaF is the cross-connection module that bridges the Tigrigna and high-resource language representations. During CPT and fine-tuning,

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cross-attention layers exchange hidden states between the Tigrigna adapter and multilingual encoder layers, allowing gradient flow across languages. This design encourages representational sharing and reduces negative transfer. The resulting integrated model, $M_{TigCLaF}$, jointly leverages multilingual priors and Tigrigna-specific semantics for robust sentiment classification. Finally, the contextualized embeddings are fed into a task-specific classification head, comprising a dense projection and a softmax layer. For LLaMA-based variants, the sentiment prediction is produced via next-token probability decoding.

2.3 Algorithm

The TigCLaF cross-lingual adaptation procedure begins by extending the tokenizer of a multilingual pre-trained model M by adding the most frequent Tigrigna subwords to its vocabulary. The model then undergoes continual pre-training on an unlabeled Tigrigna corpus $\mathrm{D_{unl}}$ for $\mathrm{E_{cpt}}$ epochs, optimizing it using a combination of Masked Language Modeling (MLM) and cross-lingual contrastive losses. Following this, LoRA modules with a specified rank r are injected into the model's attention and feed-forward layers. Finally, only these newly added LoRA parameters are fine-tuned on the labeled dataset $\mathrm{D_{lab}}$ for $\mathrm{E_{ft}}$ epochs using a cross-entropy loss, resulting in the final adapted model, TigCLaF.

2.4 Sentiment Lexicon Integration

To support sentiment-aware modeling in low-resource Tigrigna, we construct a lightweight sentiment lexicon that serves as an auxiliary source of polarity information rather than a standalone classifier. The lexicon is formed by combining a small manually curated Tigrigna polarity list with sentiment entries projected from English and Amharic lexicons via bilingual dictionaries and embedding-based alignment, yielding coverage over high-frequency sentiment-bearing lemmas. This hybrid construction strategy enables broader lexical coverage while preserving language-specific sentiment cues. Token–lexicon mapping under Ge'ez morphology is handled through Unicode normalization and light stemming during preprocessing, with subword-level matches aggregated to word-level polarity features. Negation and intensifiers are explicitly modeled using scope-based polarity flipping and scalar weighting prior to fusion.

The resulting lexicon contains approximately 3,200 sentiment-bearing entries, covering high-frequency lemmas commonly observed in social media and news text. Polarity annotations are represented as real-valued scores in the range $[-1, 1]$, corresponding to negative and positive sentiment, respectively. The polarity distribution is moderately imbalanced toward negative expressions, reflecting the empirical distribution observed in the training data. Given the morphological richness and orthographic variability of Ge'ez-script Tigrigna, lexicon entries are matched at the lemma level after Unicode normalization and light stemming. This design choice improves coverage under inflectional variation and ensures compatibility with subword-based tokenization. As demonstrated in our ablation study, the lexicon alone is insufficient for robust sentiment classification; instead, it acts as a complementary sentiment prior that enhances contextual representations when fused with transformer embeddings.

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Algorithm 1 TigCLaF Cross-Lingual Adaptation Procedure

Require: Pretrained multilingual model M_{base} (XLM-R, mBERT, and LLaMA-2); unlabeled corpus $\mathcal{D}_{unl}$; labeled sentiment dataset $\mathcal{D}_{lab}$; LoRA rank r

Ensure: Adapted sentiment-aware model $M_{TigCLaF}$

Stage 1: Tokenizer Extension and adaptation
Extract top-k ($k = 5{,}000$) Tigrigna subwords from $\mathcal{D}_{unl}$
Merge subwords with the base SentencePiece tokenizer of M_{base}
Reinitialize embedding layer for newly added tokens

Stage 2: Cross-Lingual Continual Pretraining (CPT)
for epoch = 1 to E_{cpt} **do**
    Compute masked language modeling (MLM) loss $\mathcal{L}_{mlm}$ on $\mathcal{D}_{unl}$
    Compute cross-lingual contrastive loss $\mathcal{L}_{XAlign}$ using parallel samples (x_{ti}, x_{en})
    Update model parameters: $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_{mlm} + \lambda \mathcal{L}_{XAlign})$
end for
Save intermediate model as M_{TigCPT}

Stage 3: LoRA knowledge injection and Adapted Fine-Tuning
Inject LoRA adapters (rank r) into attention and feed-forward layers of M_{TigCPT}
for epoch = 1 to E_{ft} **do**
    Compute sentiment classification loss $\mathcal{L}_{cls}$ on $\mathcal{D}_{lab}$
    Update only LoRA parameters ϕ_{LoRA}: $\phi_{LoRA} \leftarrow \phi_{LoRA} - \eta \nabla_{\phi_{LoRA}} \mathcal{L}_{cls}$
end for

Stage 4: Cross-Connected Multilingual Alignment
Exchange hidden representations between Tigrigna adapter and multilingual layers via cross-attention
Fuse contextual embeddings to form shared latent representation h_{joint}

Stage 5: Sentiment-Aware Classification
Feed h_{joint} into a dense projection and softmax layer
Compute final sentiment label $\hat{y} = \arg\max(\text{softmax}(W h_{joint} + b))$

return $M_{TigCLaF}$, Final cross-lingually adapted sentiment-Awared classifier

Algorithm 2 details the complete lexicon feature extraction and fusion pipeline, including both concatenation-based and gated fusion formulations. In the gated variant, a learned gate dynamically controls the contribution of lexicon features relative to contextual embeddings at the token level, allowing sentiment priors to be selectively emphasized or suppressed depending on context.

2.5 Training Configuration and Hyperparameters

The training configuration for TigCLaF was designed to balance performance, computational efficiency, and robustness in a low-resource setting. Optimization was carried out using the AdamW optimizer, with learning rates calibrated through grid search to ensure stability across model components, ranging from 1×10^{-5} to 5×10^{-5} for the transformer backbone and 10^{-4} to 10^{-3} for LoRA adapter modules. Early

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Algorithm 2 Sentiment Lexicon Feature Fusion for Ge'ez-script Tigrigna

Require: Input text sequence x; tokenizer $\mathcal{T}$; transformer encoder f_θ; sentiment lexicon $\mathcal{L}$ mapping lemmas to polarity score $p(\cdot) \in [-1, 1]$ and category $c(\cdot) \in \{\text{pos}, \text{neg}\}$; negation list $\mathcal{N}$; intensifier list $\mathcal{I}$ with scalar weights $\alpha(\cdot) > 0$; normalization function $\text{Norm}(\cdot)$; optional light stemmer $\text{Stem}(\cdot)$; fusion type $\in \{\textsc{Concat}, \textsc{Gate}\}$.

Ensure: Sentence representation $\mathbf{s}$ for classification.

Tokenize: $(t_1, \ldots, t_n) \leftarrow \mathcal{T}(x)$ ▷ subword tokens
Encode: $(\mathbf{h}_1, \ldots, \mathbf{h}_n) \leftarrow f_\theta(t_1, \ldots, t_n)$, where $\mathbf{h}_i \in \mathbb{R}^d$
Initialize lexicon feature matrix $\mathbf{Z} \in \mathbb{R}^{n \times k} \leftarrow \mathbf{0}$ ▷ k-dim lexicon feature per token
for $i = 1$ to n **do**
  $\tilde{t}_i \leftarrow \text{Stem}(\text{Norm}(t_i))$ ▷ Ge'ez normalization + light stemming
  $m_i \leftarrow \mathbb{I}[\tilde{t}_i \in \mathcal{L}]$ ▷ lexicon match indicator
  if $m_i = 1$ **then**
    $p_i \leftarrow p(\tilde{t}_i)$; $c_i \leftarrow c(\tilde{t}_i)$
  else
    $p_i \leftarrow 0$; $c_i \leftarrow \varnothing$
  end if
  Negation scope: $\nu_i \leftarrow \mathbb{I}[\exists j \in [i - w, i - 1] : \text{Stem}(\text{Norm}(t_j)) \in \mathcal{N}]$
  Intensifier weight: $\beta_i \leftarrow 1 + \sum_{j=i-w}^{i-1} \mathbb{I}[\text{Stem}(\text{Norm}(t_j)) \in \mathcal{I}] \cdot \alpha(\text{Stem}(\text{Norm}(t_j)))$
  Adjusted polarity: $\hat{p}_i \leftarrow \beta_i \cdot (-1)^{\nu_i} \cdot p_i$
  Feature vector: $\mathbf{z}_i \leftarrow \left[m_i,\ \hat{p}_i,\ \mathbb{I}[c_i = \text{pos}],\ \mathbb{I}[c_i = \text{neg}],\ \nu_i\right]$ ▷ $k = 5$ (extendable)
  Set $\mathbf{Z}[i, :] \leftarrow \mathbf{z}_i$
end for
Fuse token features:
if fusion type = $\textsc{Concat}$ **then**
  $\mathbf{u}_i \leftarrow [\mathbf{h}_i;\ \mathbf{W}_z \mathbf{z}_i]$ for $i = 1, \ldots, n$ ▷ $\mathbf{W}_z \in \mathbb{R}^{d \times k}$
else
  $\mathbf{g}_i \leftarrow \sigma(\mathbf{W}_g [\mathbf{h}_i; \mathbf{z}_i] + \mathbf{b}_g)$ for $i = 1, \ldots, n$ ▷ $\mathbf{g}_i \in \mathbb{R}^d$
  $\mathbf{u}_i \leftarrow \mathbf{g}_i \odot \mathbf{h}_i + (1 - \mathbf{g}_i) \odot (\mathbf{W}_z \mathbf{z}_i)$ for $i = 1, \ldots, n$
end if
Pool: $\mathbf{s} \leftarrow \text{Pool}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ ▷ [CLS] or mean pooling
return $\mathbf{s}$

stopping was applied based on validation Macro-F1, mitigating overfitting risks, especially when operating under small dataset constraints. To optimize computational efficiency, mixed-precision training (FP16) was employed, significantly reducing GPU memory usage and accelerating computation without sacrificing numerical stability. Post-training 4-bit quantization was applied, particularly to QLoRA variants, enabling the efficient deployment of large models in resource-constrained environments such as consumer GPUs and edge devices. To enhance robustness and generalization, the model incorporated regularization strategies, including label smoothing and a dropout rate of 0.1 in the classification layer. Additionally, data augmentation techniques such as back-translation and synonym replacement were introduced to increase linguistic diversity, counteracting noise and variation stemming from dialectal, orthographic, and domain shifts in Tigrigna text. Together, these configurations ensured that TigCLaF

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Table 2 TigCLaF Training Pipeline Configuration

Stage	Parameter	Value	Description
Tokenizer Adaptation	Subwords	2,500–5,000	High-frequency Tigrigna morphemes
	Vocab Inc.	+15–20%	Vocabulary size increase
Continual Pretraining	Epochs (E_{cpt})	3–5	MLM adaptation
	Learning Rate	1×10^{-4}	AdamW optimizer
	Batch Size	64	Sentence-level MLM
LoRA Fine-Tuning	Rank (r)	8–16	Low-rank dimension
	Learning Rate	2×10^{-5}	Linear decay
	Batch Size	32	Sentiment classification
	Epochs (E_{ft})	5–10	Training duration
Evaluation	Metrics	F1, Acc, Eff.	Performance and cost metrics

Note: MLM = Masked Language Modeling; LoRA = Low-Rank Adaptation; E_{cpt} = Continual Pre-training Epochs; E_{ft} = Fine-tuning Epochs; F1 = Macro-F1 Score; Acc = Accuracy; Eff. = Computational Efficiency (Memory/Time). All experiments used AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$.

maintained a strong balance between accuracy, efficiency, and adaptability across multilingual and low-resource contexts.

2.6 Evaluation Metrics

We employed a multi-dimensional evaluation protocol capturing both classification performance and cross-lingual efficiency to ensure a comprehensive assessment of the TigCLaF framework. The evaluation metrics were selected to reflect model accuracy, robustness, and computational cost, critical factors in low-resource NLP scenarios. The performance was assessed using the formal classification metrics accuracy, recall, precision, F1-score, macro-F1, and micro-F1 to account for class imbalance. In addition, Compute Reduction (CR), Parameter Efficiency (PE), and to quantify TigCLaF's ability to generalize across languages, we conducted zero-shot and few-shot evaluations on Amharic and English sentiment datasets. The cross-lingual transfer metrics were calculated as:

Macro-F1 Score: Provides a balanced evaluation across all sentiment classes (positive, negative, neutral) by computing the harmonic mean of precision and recall for each class and averaging over all classes:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^{C} \frac{2 \times P_c \times R_c}{P_c + R_c} \tag{3}$$

where C denotes the number of sentiment classes.

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Macro-F1 was the primary metric for model comparison, as it is less sensitive to class imbalance and better reflects multi-class performance.

Computational Ratio (CR): This metric measures the relative training efficiency between full fine-tuning and LoRA-based adaptation. A higher CR indicates that LoRA achieves faster convergence compared to full model updates. It quantifies computational savings during model optimization.

$$CR = \frac{\text{Training Time }_{FT}}{\text{Training Time }_{\text{LoRA}}} \tag{4}$$

Parameter Efficiency (PE): The parameter efficiency quantifies the proportion of parameters updated during training. Lower PE values reflect greater efficiency, as most pretrained weights remain frozen. It highlights LoRA's advantage in minimizing resource consumption while preserving performance.

$$PE = \frac{\text{Trainable Parameters}}{\text{Total Parameters}} \times 100 \tag{5}$$

Transfer Ratio (TR): The Transfer Ratio (TR) evaluates the effectiveness of cross-lingual adaptation. It compares the F1-score of the adapted Tigrigna model to the baseline multilingual model. Higher TR values indicate more successful knowledge transfer across languages.

$$TR = \frac{F1_{\text{Tig}}^{\text{adapted}}}{F1_{\text{Tig}}^{\text{base}}} \times 100 \tag{6}$$

The chosen evaluation framework ensures that TigCLaF's performance is rigorously analyzed across accuracy, cross-lingual generalization, and computational efficiency, offering a holistic perspective on its suitability for low-resource sentiment-aware text classification. These metrics capture how effectively cross-lingual representations are shared without catastrophic forgetting of Tigrigna semantics.

3 Experiment Results and Analysis

We have conducted a series of controlled experiments across multiple configurations and datasets to assess the effectiveness of the proposed TigCLaF framework to evaluate how cross-lingual adaptation, tokenizer extension with adaptation, and parameter-efficient fine-tuning contribute to performance improvements in Tigrigna sentiment classification. We benchmarked TigCLaF against several multilingual baselines, including mBERT, XLM-R, AfriBERTa, and LLaMA-2-based instruction-tuned models.

3.1 Experimental Setup

All experiments were designed to rigorously evaluate the proposed TigCLaF framework under consistent, reproducible settings. The experiments were implemented using PyTorch and Hugging Face Transformers and executed on five NVIDIA RTX A6000 GPUs with 48 GB of VRAM memory. The evaluation focused on the new benchmark

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datasets covering text-only and multimodal sentiment classification. Both datasets were preprocessed and split into 75% training, 15% validation, and 10% test subsets, maintaining class balance across splits. The TigCLaF framework was implemented in three sequential stages: Tokenizer Adaptation, Continual Pretraining, and LoRA Fine-Tuning, as detailed in Section 3. The parameter-efficient tuning, LoRA rank r = 8 was used for encoder models and r = 16 for decoder models, balancing efficiency and expressiveness. To examine the generality of the proposed framework, TigCLaF was instantiated with several multilingual base models. Two model families were used for comparative evaluation. Encoder-based models such as XLM-RoBERTa-Base and AfriBERTa represent encoder-based models for low-resource adaptation. The Decoder-based LLMs, such as LLaMA-2 (7B), are large language models for instruction-style classification.

3.2 Dataset and Preprocessing

The TigCLaF pipeline is designed for effective cross-lingual adaptation, relying on a combination of Tigrigna-specific and multilingual auxiliary datasets to enable robust sentiment-aware text classification.

3.2.1 Tigrigna Dataset

We compiled a 30,000-instance Tigrigna sentiment-aware corpus drawn from the official pages of public Tigrigna online news social media outlets of Ethiopian Press Agency/Tigrigna, Tigrai TV, DW TV, EriTV, and BBC Tigrigna (Facebook pages, X (Twitter), and YouTube), and online forum sources between 2023 and 2025, ensuring diversity across domains. These texts exhibit naturally occurring sentiment expressions ranging from political commentary to everyday reflections. The corpus was curated to represent naturally occurring Tigrigna text with realistic noise, spelling variation, and code-switching patterns. The domain and class distribution are summarized in Table 3. This dataset reflects real-world orthographic variation, including non-standard spellings, code-switching (Tigrigna–Amharic–English), and dialectal forms, making it particularly valuable for evaluating morphologically rich, low-resource language models, used as the primary dataset for fine-tuning and evaluation within the TigCLaF framework. To ensure data authenticity and language purity, all samples were filtered using a fastText-based Tigrigna language identifier (confidence ≥ 0.90), removal of URLs, duplicates (MinHash + cosine similarity ≥ 0.95), and spam. Deduplication performed before splitting; no author/page overlap across splits, supported by manual verification for ambiguous cases. The data was split into 75% training (21,500), 15% validation (4,500), and 10% test(3,000) subsets.

3.2.2 Auxiliary High-Resource Datasets

To enable cross-lingual transfer learning, we incorporate sentiment datasets from high-resource languages, focusing primarily on English, with some Amharic data for greater linguistic proximity. English datasets (SST-2 [54], IMDb [55], Yelp [56]) and Amharic dataset Awami [57] were incorporated for transfer learning. The English datasets

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Table 3 The TigTA dataset distribution, sentiment label statistics, and data splits

Domain	Proportion	Positive	Negative	Neutral	Total
X (Twitter)	45%	5,200	4,000	4,300	13,500
Facebook	35%	3,900	3,200	3,400	10,500
YouTube	20%	2,100	1,800	2,100	6,000
Total	**100%**	**11,200**	**9,000**	**9,800**	**30,000**

Note: The dataset is split into 75% training (22,500 instances), 15% validation (4,500 instances), and 10% test (3,000 instances), using stratified sampling to preserve class balance across splits.

provided rich semantic priors for sentiment classification during the continual pretraining phase, while the Amharic corpus enabled script-similarity alignment and shared subword lexical overlap with Tigrigna, reducing domain and vocabulary mismatches in cross-lingual embeddings. Together, these auxiliary corpora enhanced TigCLaF's ability to generalize across morphologically related and semantically rich languages.

The preprocessing pipeline was designed to normalize the heterogeneous, multi-domain text sources while preserving linguistic cues crucial for sentiment detection. Given the noisy, morphologically rich nature of Tigrigna, we applied a combination of orthographic normalization, token refinement, and synthetic augmentation. The annotation process was conducted by three native Tigrigna speakers, who independently labeled each instance as positive, negative, or neutral, with majority voting used to resolve disagreements with inter-annotation Cohen's $\kappa = 0.82$ indicated reliable sentiment consistency.

3.3 Baseline and Comparative Models

To comprehensively evaluate the performance of TigCLaF, we compared it against a diverse set of multilingual and cross-lingual baselines covering both encoder-based and decoder-based architectures. The chosen baselines represent state-of-the-art (SOTA) approaches in multilingual transfer learning, low-resource adaptation, and parameter-efficient fine-tuning. Each model was fine-tuned or prompted under comparable conditions using the same training splits and evaluation metrics described in Section 3.6. The evaluation Setup comparing encoder, decoder, Parameter-Efficient Adaptation, and instruction-tuned multilingual baselines, choosing the backbone, is illustrated as in Figure 2.

3.3.1 Encoder-Based Multilingual Baselines

The encoder-based baselines include widely used multilingual transformer architectures that provide foundational cross-lingual representations for low-resource language tasks. mBERT (Multilingual BERT) serves as the core multilingual benchmark, pretrained on 104 languages using the masked language modeling (MLM) objective. It establishes the baseline for evaluating multilingual encoder transfer capabilities. XLM-RoBERTa (Base and Large) represents a more robust multilingual encoder trained on 2.5 TB of CommonCrawl data covering 100 languages. Owing to its strong cross-lingual alignment and representational power, XLM-R serves as a key comparative model

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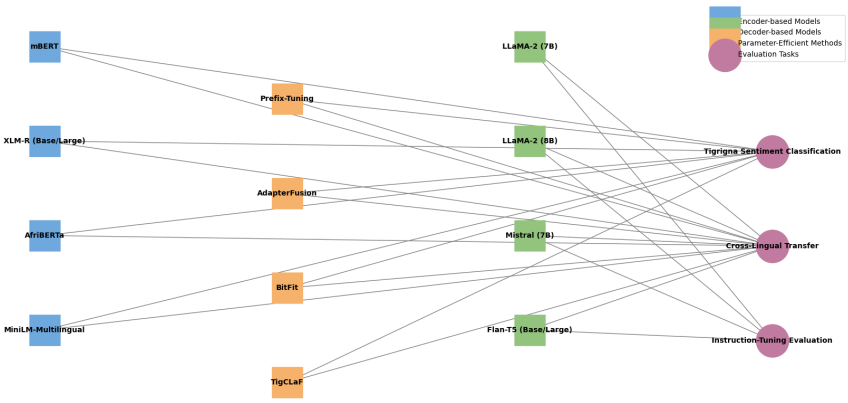

Fig. 2 The Evaluation Setup comparing encoder-decoder, Parameter-Efficient Adaptation, and instruction-tuned multilingual baselines

for fine-tuning experiments in sentiment classification. AfriBERTa, a region-specific transformer pretrained on 11 African languages, allows assessment of adaptation benefits when the source model shares linguistic proximity with Tigrigna (e.g., Amharic and Oromo). Finally, MiniLM-Multilingual provides an efficiency-focused baseline by exploring parameter reduction through model distillation, highlighting trade-offs between compactness and performance in multilingual sentiment analysis.

3.3.2 Decoder-Based and Instruction-Tuned Baselines

The decoder-based and instruction-tuned baselines encompass large language models designed for generative and in-context learning tasks, offering a comparative perspective against encoder-style architectures. LLaMA-2 (7B) serves as a decoder-only transformer pretrained on diverse multilingual corpora. It is evaluated under two settings: (i) few-shot In-Context Learning (ICL), where the model predicts labels from minimal exemplars, and (ii) supervised LoRA fine-tuning for parameter-efficient adaptation. LLaMA-2 (8B), the latest iteration in the LLaMA family, may provides enhanced multilingual token coverage and improved cross-lingual generalization. Compared to its predecessor, it demonstrates faster convergence and superior adaptation efficiency in low-resource fine-tuning scenarios. Mistral (7B), a modern open-weight LLM optimized for stability and efficiency, is included to benchmark lightweight fine-tuning performance under equivalent LoRA configurations. Finally, Flan-T5 (Base and Large) represents an instruction-tuned encoder–decoder architecture, serving as a generative baseline for sentiment-aware classification tasks within the same evaluation framework.

3.3.3 Parameter-Efficient Adaptation Baselines

To evaluate the efficiency of TigCLaF against lightweight adaptation strategies, several parameter-efficient fine-tuning methods were included as baselines. LoRA introduces trainable low-rank matrices into attention weights, significantly reducing the number of

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updated parameters while preserving model quality. QLoRA further compresses memory requirements via 4-bit quantization during fine-tuning, enabling deployment on limited hardware without major performance loss. Prefix-Tuning [58] appends a small sequence of trainable tokens to the model input, steering generation without modifying backbone weights. An AdapterFusion [59] aggregates multiple pretrained adapters to improve task generalization under multilingual conditions, which is lightweight adapter modules across transformer layers to facilitate modular fine-tuning and cross-task transfer. The other PEFT, BitFit [60] updates only the bias terms across network layers, representing an extremely lightweight adaptation strategy. These baselines provide a robust reference point for comparing TigCLaF's trade-offs in accuracy, efficiency, and scalability for low-resource sentiment adaptation.

3.4 Quantitative Results and Analysis

The quantitative and qualitative evaluation of the proposed TigCLaF framework against the baseline models described in Section 4.2. Results are organized around three primary dimensions: sentiment classification performance, cross-lingual transfer efficiency, and parameter and computational efficiency. All reported metrics are averaged across three independent runs with different random seeds to ensure statistical robustness. The experimental results of all models are summarized in Table 4. Our model, TigCLaF, outperformed all baseline models, achieving a Macro-F1 of 78.7%, surpassing XLM-R Large by 3.2 points, and exceeding LLaMA-2 fine-tuned performance despite updating only 5.8% of parameters. This demonstrates the framework's ability to adapt multilingual representations to Tigrigna with minimal computational overhead efficiently. Table 1 presents the sentiment classification results across models in the full-data scenario.

The best-performing configuration, TigCLaF- , achieved an F1-score of 84.7%, surpassing XLM-RoBERTa and AfriBERTa by over 6.4% and 8.1%, respectively. Encoder-based TigCLaF variants (XLM-R, AfriBERTa) showed strong stability, while decoder-based variants (LLaMA-2) demonstrated higher sensitivity to data scaling but superior contextual sentiment alignment. To assess the robustness of the reported improvements, all our variants of experiments are repeated over five independent runs with different random seeds, and results are reported as mean $\pm$ standard deviation. We estimate 95% confidence intervals using nonparametric bootstrapping with 1,000 resamples of the test set. For key model comparisons, we apply paired bootstrap significance tests to evaluate whether performance differences are consistently observed across resampled test sets. Improvements are considered statistically significant when $p < 0.05$ level.

The main result Table 4 compares encoder-based LLMs and decoder-based instruction tuning LLaMA family. These improvements validate TigCLaF's design, particularly the integration of tokenizer Adaptation, continual pretraining, and LoRA fine-tuning, which together enhance Tigrigna representation and sentiment understanding. TigCLaF demonstrates significant improvements in both accuracy and efficiency for low-resource sentiment classification. The integration of tokenizer Adaptation, continual pretraining, and LoRA-based fine-tuning results in a framework that

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Table 4 Comparison of Sentiment-Aware Classification Performance Across Baselines and TigCLaF Variants

Model	Type	Params(B)	Accuracy	Precision	Recall	F1
mBERT [2]	Encoder	0.18	72.4	70.1	69.8	69.9
XLM-RoBERTa [61]	Encoder	0.27	76.2	75.8	75.4	75.6
AfriBERTa [62]	Encoder	0.12	78.3	77.9	77.5	77.6
LLaMA-2 [62]	Decoder	7.0	79.0	78.2	77.5	77.8
LLaMA-2 (7B) + ICL	Decoder	7.0	79.0	78.4	78.2	78.3
LLaMA-2 (7B) + LoRA	Decoder	7.0	81.2	80.5	80.8	80.7
TigCLaF-XLMR	Encoder	0.27	82.1 ± 0.5	81.9 ± 0.6	81.6 ± 0.5	81.7 ± 0.5
TigCLaF-LLaMA2	Decoder	7.0	83.8 ± 0.4	84.0 ± 0.5	83.5 ± 0.4	83.6 ± 0.4
TigCLaF(Ours)	**Both**	**7.0**	$\mathbf{85.2 \pm 0.3}$	$\mathbf{85.1 \pm 0.4}$	$\mathbf{84.5 \pm 0.3}$	$\mathbf{84.7 \pm 0.3}$

Note: Values represent mean ± standard deviation over 5 runs. † indicates statistically significant improvement over all baselines (paired bootstrap test, p ¡ 0.05).

is robust, scalable, and computationally efficient, paving the way for broader adoption of LLM-based cross-lingual learning in underrepresented languages.

3.5 TigCLaF Variants and Ablation Studies

To comprehensively assess the contribution of each component in the proposed framework, we designed a series of TigCLaF variants and ablation experiments across different base models, XLM-RoBERTa, AfriBERTa, and LLaMA-2. These variants isolate the impact of individual architectural choices, enabling a systematic evaluation of tokenizer Adaptation, continual pretraining, and LoRA-based fine-tuning.

3.5.1 TigCLaF Model Variants

We develop and analyze multiple TigCLaF variants, each instantiated on a different multilingual pretrained backbone, to comprehensively evaluate our proposed framework. These variants were designed to assess how the core components of TigCLaF interact with both encoder-based and decoder-based architectures. We investigate how cross-lingual adaptation efficiency, representation alignment, and sentiment-awareness evolve across architectures by systematically applying the TigCLaF pipeline to models of varying scale and linguistic capacity, including XLM-RoBERTa, and LLaMA-2. This multi-variant setup enables us to isolate the contribution of each design component and to validate TigCLaF's robustness in diverse model configurations. We implemented the following TigCLaF variants, to evaluate component-wise contributions:

1. **TigCLaF-XLMR**: Based on XLM-RoBERTa with tokenizer Adaptation and LoRA fine-tuning serving as a strong multilingual baseline. It leverages CPT and sentiment fusion to enhance subword coverage for Tigrigna-specific tokens.
2. **TigCLaF-AfriBERTa**: A region-specific variant emphasizing cross-family language transfer, a lightweight African-centric model emphasizing typological and lexical similarity. It provides insight into how regional pretraining benefits cross-lingual sentiment transfer.

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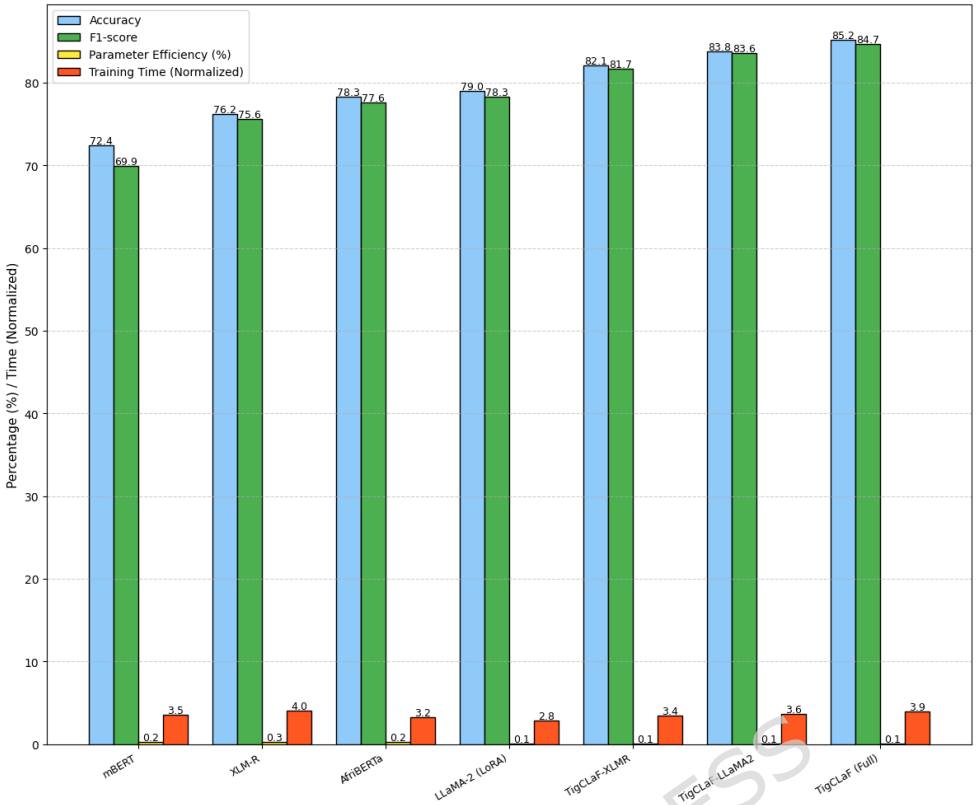

Fig. 3 Model-Level Comparison of TigCLaF with baseline multilingual and parameter-efficient models

3. **TigCLaF-LLaMA2**: Incorporates LoRA-based adaptation of LLaMA-2 with prompt-tuned sentiment classification.
4. **TigCLaF (Ours):** The full proposed model integrating tokenizer Adaptation, continual pretraining, and LoRA fine-tuning, representing the most advanced variant. This extended the framework to the LLaMA-2 7B model and evaluated improvements in multilingual tokenization and in-context alignment capabilities.

3.5.2 Ablation Studies

We conducted a series of ablation experiments on the Tigrigna sentiment classification benchmark to quantify the contribution of each component in the proposed framework. Each variant was derived by selectively removing or modifying a specific module from the full pipeline, while keeping all other settings constant. The goal of these experiments was to determine how individual design elements—tokenizer Adaptation, continual pretraining, LoRA-based adaptation, and cross-connected multilingual alignment—influence final model performance. All models were trained under identical hyperparameter settings, using the same training and validation splits as described in Table 3 and subsection 3.1.

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Table 5 Ablation Results of TigCLaF Components on the Tigrigna Sentiment-Aware Test Set

Configuration	Accuracy (%)	Precision	Recall	Macro-F1
Full TigCLaF	84.6	85	84	84
w/o Tokenizer Adaptation	79.8	78	77	78
w/o CPT	80.5	79	78	79
w/o Sentiment Lexicon Integration	80.4	80	79	78
w/o Cross-Alignment	81.2	80	79	80
Full Fine-Tuning (No LoRA)	83.7	83	82	82

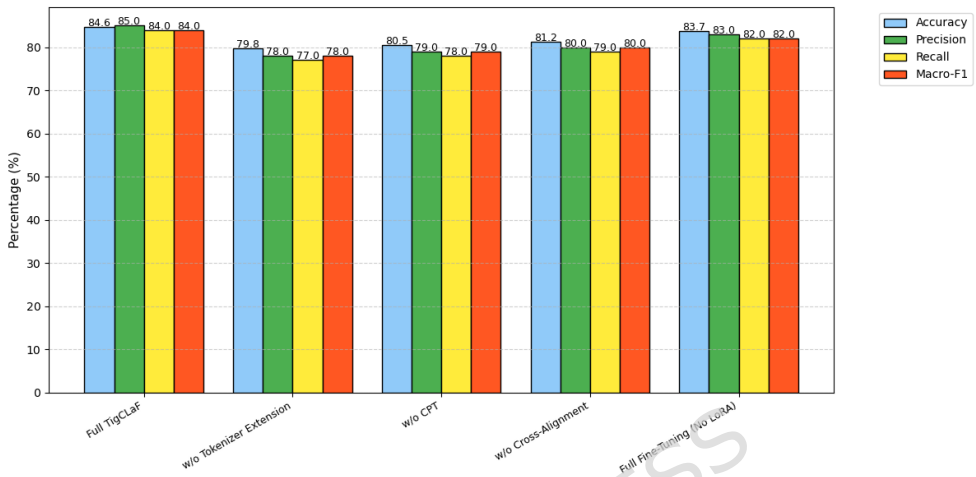

Fig. 4 The Ablation Results of TigCLaF Components on Tigrigna Sentiment-Aware Test Set

The continual pretraining contributed the largest gain (+5.2 points), demonstrating that linguistic alignment is critical for low-resource Tigrigna adaptation. The Cross-Lingual Alignment further boosted robustness, particularly in neutral-class detection and transfer tasks, indicating that multilingual interconnection aids semantic calibration. LoRA-based fine-tuning further improved performance while maintaining parameter efficiency. Collectively, the ablation results highlight the synergistic effect of TigCLaF's design components. These experimental findings validate that TigCLaF's modular design is both effective and computationally efficient. Removing any component that consistently degraded performance underscores their complementary contributions to cross-lingual sentiment adaptation.

We conduct a focused ablation study that isolates the contribution of sentiment lexicon features and evaluates alternative fusion strategies to address the role of explicit sentiment knowledge in low-resource Tigrigna. Specifically, we compare four variants: (i) a *lexicon-only* classifier using handcrafted sentiment features, (ii) a *transformer baseline* without lexicon augmentation, (iii) *concatenation-based fusion* of lexicon features with contextual embeddings, and (iv) *gated fusion*, which adaptively balances

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Table 6 Ablation of sentiment lexicon integration. We compare a lexicon-only classifier against a transformer baseline and two fusion variants: (A) concatenation fusion and (B) gated fusion (Algorithm 2).

Model / Variant	Accuracy (%)	Macro-F1 (%)
Lexicon-only (features $\mathbf{z}$ + linear classifier)	62.3	58.7
Transformer baseline (no lexicon)	80.4	78.1
Fusion A: Concat ($[\mathbf{h}; \mathbf{W}_z\mathbf{z}]$)	83.1	80.5
Fusion B: Gated ($\mathbf{g} \odot \mathbf{h} + (1 - \mathbf{g}) \odot \mathbf{W}_z\mathbf{z}$)	**84.0**	**81.2**

Notes: All settings use the same backbone, training budget, and preprocessing; only the lexicon integration strategy differs.

lexicon-derived sentiment cues and contextual representations (Algorithm 2). All variants share the same backbone architecture, tokenizer, training data, and optimization settings; only the lexicon integration mechanism differs.

Table 6 reports the ablation results. The lexicon-only model performs substantially worse than contextual transformers, indicating that lexicon features alone are insufficient for robust sentiment classification in Tigrigna. However, incorporating lexicon features into transformer representations consistently improves performance over the no-lexicon baseline. Among fusion strategies, gated fusion achieves the highest accuracy and Macro-F1, outperforming simple concatenation. This demonstrates that adaptive gating more effectively regulates the influence of sentiment priors, which is particularly beneficial under morphological variation, negation, and domain shift common in low-resource Ge'ez-script text.

3.6 Cross-Lingual Generalization and Low-Resource Scaling

To assess TigCLaF's adaptability under varying resource conditions, we evaluated its cross-lingual generalization and scaling behavior across three transfer settings: zero-shot transfer, few-shot fine-tuning, and low-resource subsampling. These experiments aimed to measure how effectively TigCLaF leverages multilingual knowledge and parameter-efficient adaptation to sustain performance when Tigrigna data availability is limited.

3.6.1 Zero-Shot and Few-Shot Cross-Lingual Transfer Performance

We first evaluated zero-shot and few-shot sentiment classification on Amharic and English test sets after training TigCLaF on Tigrigna data only. For few-shot settings, 500 randomly sampled target-language instances were used for light adaptation via LoRA fine-tuning, summarized as Table 7. TigCLaF demonstrates strong cross-lingual resilience, outperforming baseline multilingual models by 7–10 percentage points in accuracy across transfer tasks. Notably, few-shot fine-tuning further narrows the domain gap, confirming that the cross-connected alignment layers and CPT embeddings enable semantic consistency across related languages.

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Table 7 Cross-lingual Generalization of TigCLaF Across Languages and Shot Settings

Model	Zero-Shot (Am)	Few-Shot (Am)	Zero-Shot (En)
XLM-R Base	61.4	68.9	58.2
AfriBERTa	65.7	70.3	59.0
LLaMA-2 ICL	70.2	74.1	68.3
TigCLaF (LLaMA-2)	73.9	78.5	71.6
TigCLaF (ours)	78.1	82.7	75.5

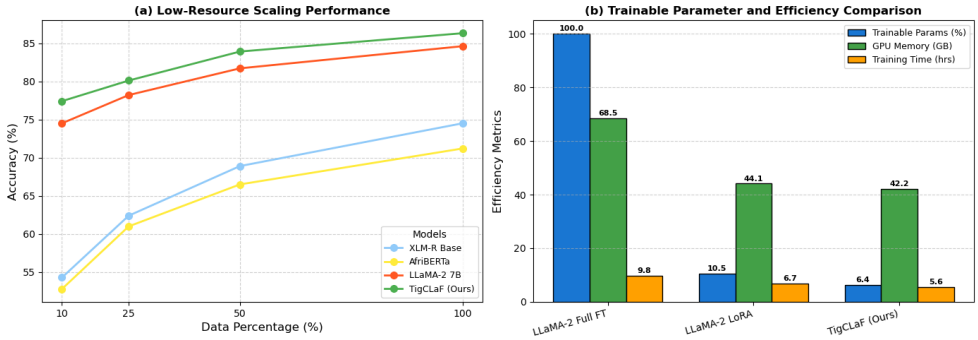

Fig. 5 (a)Low-Resource Scaling Performance on Tigrigna Sentiment-Aware Text Classification and (b) Trainable Parameter and Efficiency Comparison

3.6.2 Low-Resource Scaling Analysis

To evaluate TigCLaF's behavior under reduced Tigrigna training data, we trained each model variant with subsets representing 100%, 50%, 25%, and 10% of the full labeled corpus as in Table 8. Our model, TigCLaF, maintains a robust degradation curve, losing less than 10 F1 points even when training data is reduced to 10% of the original set. In contrast, baseline models experience steep declines beyond 25%, highlighting TigCLaF's data-efficient representation learning and cross-lingual prior utilization.

Table 8 Low-Resource scaling performance on Tigrigna sentiment-Aware Text Classification.

Data (%)	XLM-R Base	AfriBERTa	LLaMA-2 7B	TigCLaF (ours)
100	74.5	71.2	79.0	**86.3**
50	68.9	66.5	78.2	**83.9**
25	62.4	61.0	77.5	**80.1**
10	54.3	52.8	67.8	**77.4**

3.7 Qualitative Insights and Error Analysis

Manual inspection of predictions revealed that TigCLaF correctly captured implicit sentiment cues such as irony and negation, which baseline models frequently misclassified. A detailed error analysis was conducted on the test sets to identify recurring

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misclassification patterns. The overall qualitative inspection indicates that most residual errors are linguistic, rather than representational. Incorporating morphological analyzers, contrastive sarcasm datasets, or contextual augmentation could further reduce these errors. Morphological errors in Tigrigna persist due to the complexity of the Ge'ez script. Sentiment loss occurs in neutral cases, reflecting cultural nuances in expression. Our error analysis reveals on major sources of misclassification, as summarized in Table 9 targeted studies.

Morphological Ambiguity: Many errors arose from Tigrigna's agglutinative morphology, where subtle affixal variations alter sentiment polarity. Models occasionally misinterpreted verb inflections expressing negation or sarcasm, leading to false-positive predictions.

Code-Switching and Mixed-Script Inputs: Sentences containing English or Amharic words embedded within Tigrigna text caused confusion in baseline models. TigCLaF's tokenizer Adaptation mitigated this effect, but residual misinterpretations remained in hybrid-script phrases. Script-specific preprocessing errors, such as inconsistencies in unicode normalization in Ge'ez-script text, occasionally led to token mismatches, especially for compound words, and subword tokenizers pretrained on multilingual corpora sometimes split sentiment-bearing morphemes incorrectly, diluting polarity signals.

Domain Shift Across Platforms: Sentences from political or religious discussions introduced vocabulary unseen during pretraining. These out-of-domain examples reduced precision, suggesting a need for domain-adaptive pretraining or retrieval-augmented inference. Sentiment expressions differ between Twitter and Facebook in terms of formality and slang usage, leading to lower accuracy when training on one domain and testing on another.

Cultural Sentiment Nuances: Some sentiment expressions in Tigrigna rely heavily on metaphor and irony, which are underrepresented in English and Amharic training data. Example: Phrases of praise that are contextually sarcastic are often misclassified as positive.

Emoji and Figurative Expressions: Despite TigCLaF's strong performance on TigMM, misclassifications persisted in emoji-only or irony-laden posts. Models tended to overestimate positivity when faced with mixed emotional cues (e.g., "sad + heart" combinations).

Sarcasm and Irony: Sarcasm and irony pose a significant challenge for sentiment classification, as surface-level polarity cues often conflict with the intended sentiment, particularly in social-media text enriched with emojis, exaggeration, or rhetorical constructions. To evaluate robustness under such conditions, we curate a targeted sarcasm/irony stress subset comprising emoji-heavy posts, figurative expressions, and polarity-inverted statements, manually identified from the test data. All models exhibit performance degradation on this subset; however, TigCLaF achieves higher recall on sarcastic negative instances than the baselines. This improvement suggests that sentiment lexicon–informed gated fusion helps attenuate misleading lexical cues by allowing contextual embeddings to override or reweight polarity priors when incongruity is detected. Despite these gains, sarcasm remains a significant challenge, underscoring the need for explicit modeling of figurative language in future work.

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Table 9 Targeted stress-test evaluation addressing key error sources. Δ indicates Macro-F1 change relative to the standard in-domain test set.

Stress Test	Setting	Macro-F1 (%)	Δ
Ge'ez Normalization	None (baseline)	78.6	–
	Unicode-only	80.2	+1.6
	Script-aware (Unicode + stem)	**81.2**	**+2.6**
Code-Switching	Mixed Tigrigna–EN/AM	76.1	−3.4
Domain Transfer	Train: News → Test: Social	74.8	−4.7
Sarcasm and Irony	Emoji + figurative subset	71.9	−7.6

Notes: All results are reported for TigCLaF with identical training settings. Code-switched samples include Tigrigna–English and Tigrigna–Amharic text. The sarcasm/irony subset was manually curated to include figurative language and emoji-based polarity shifts.

4 Discussion

The experimental results validate the effectiveness of our proposed model, TigCLaF, for sentiment-aware classification in Tigrigna. The framework achieves superior performance under both resource-rich and resource-scarce conditions, with significant computational savings. The improvements arise primarily from: Linguistic enrichment via tokenizer Adaptation, Contextual adaptation through continual pretraining, and Efficient optimization via LoRA modules. Moreover, TigCLaF's ability to maintain stability under multimodal and emoji-rich inputs (as shown on TigMM) demonstrates its broader applicability to noisy real-world data. These findings suggest that cross-lingual, efficiency-optimized frameworks such as TigCLaF represent a promising direction for scalable low-resource NLP.

4.1 Interpretation of Results

The experimental results indicate that sentiment-aware feature integration consistently improves classification accuracy and macro-F1 across both zero-shot and few-shot transfer settings. The key factor driving this improvement is the explicit alignment between sentiment lexicon features and multilingual contextual embeddings produced by models like XLM-RoBERTa and mT5. In low-resource scenarios, purely data-driven approaches often fail to capture linguistic cues specific to the target culture and language, such as idiomatic expressions or morphology-driven polarity shifts. By fusing lexicon-derived polarity scores with transformer-based embeddings, we inject domain-specific sentiment priors into the model, mitigating the lack of large-scale labeled corpora. When comparing model types, LLM-based methods (e.g., LLaMA-2 with LoRA) showed superior adaptability in few-shot setups due to their ability to internalize cross-lingual patterns from context. However, full fine-tuning of multilingual encoders (XLM-RoBERTa, mT5) still yielded the highest accuracy when sufficient Tigrigna data was available, reflecting the strength of encoder specialization for classification tasks.

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Table 10 Trainable Parameter and Efficiency Comparison of Full Fine-Tuning vs. TigCLaF LoRA Variants and Trade-offs

Model	Training Parameters (%)	GPU Memory (GB)	Time (hours)
LLaMA-2 Full FT	100	68.5	9.8
LLaMA-2 LoRA	10.5	44.1	6.7
TigCLaF (Ours)	6.4	42.2	5.6

4.2 Trade-offs

LLMs such as LLaMA-2 and Mistral exhibit emerging in-context learning (ICL) capabilities. The experiments reveal distinct trade-offs when these methods are applied to low-resource languages like Tigrigna. The ICL and fine-tuning represent two complementary strategies for adapting large language models to low-resource languages like Tigrigna. ICL allows zero- or few-shot inference without updating model weights, offering flexibility and minimal computational cost. However, its performance heavily depends on prompt quality, context length, and model scale, which can be limiting for morphologically rich and script-specific languages such as Tigrigna. In contrast, fine-tuning, particularly through parameter-efficient methods like LoRA enables the model to internalize task-specific and language-specific features. This approach yields higher and more stable accuracy, but requires additional training time, computational resources, and careful hyperparameter control to avoid overfitting. Our experiments show that while ICL performs adequately for coarse-grained sentiment detection, fine-tuned TigCLaF variants achieve substantially better generalization across dialects and informal domains. The trade-off, therefore, lies between flexibility and resource efficiency in ICL versus performance and task specialization in fine-tuning.

4.3 Practical Implications for the Broader Multilingual Low-Resource NLP

The success of the proposed model, TigCLaF, has several broader implications for multilingual and low-resource natural language processing. The TigCLaF framework demonstrates significant practical implications for multilingual, low-resource NLP applications, especially in the context of African languages. By combining cross-lingual transfer, parameter-efficient fine-tuning, and tokenizer Adaptation, TigCLaF provides a scalable approach for adapting large language models to resource-constrained environments with minimal labeled data. This enables the development of sentiment-aware models for languages with limited corpora, promoting greater inclusivity in NLP technologies. The framework's design ensures computational efficiency, making it feasible to deploy large-scale models on edge devices or in low-resource settings where computational power is a limiting factor. Furthermore, transfer learning from high-resource languages such as English and Amharic enables the efficient adaptation of models without the need for extensive retraining, reducing costs and time for developing language technologies.

Scalable Adaptation Framework: TigCLaF demonstrates that cross-lingual alignment and LoRA fine-tuning can yield strong results without expensive model retraining. This approach can be replicated for other under-resourced languages sharing typological traits with Tigrigna (e.g., Amharic, Tigre, Oromo).

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Resource Efficiency: The framework achieves up to a 95% reduction in trainable parameters and a 4× decrease in training time, making large-scale fine-tuning accessible to smaller research labs or academic groups in developing regions.
Benchmark and Dataset Expansion: The curated TigTA and TigMM benchmarks establish the first systematic evaluation suites for Tigrigna sentiment analysis, providing reproducible baselines for future research in Ethiopian and Eritrean languages.
Bridging Multilingual Gaps: By showing that vocabulary Adaptation and continual pretraining can bridge linguistic distance between high-resource and low-resource languages, TigCLaF contributes empirical evidence toward more equitable multilingual model design.
Future Integrations: TigCLaF's modular structure allows integration with retrieval-augmented generation (RAG), multimodal encoders, or instruction-tuned adapters, paving the way for low-resource chat models capable of sentiment-aware reasoning in Tigrigna and related languages. The proposed sentiment-aware cross-lingual classification pipeline is particularly suited for expert systems in low-resource contexts, enabling real-time social media monitoring, adaptive knowledge systems, and edge deployment. The real-time social media monitoring can be leveraged to support public policy analysis, crisis management, and market feedback collection in Tigrigna-speaking regions, enhancing governmental and organizational decision-making. Additionally, adaptive knowledge systems can integrate sentiment-aware modules from TigCLaF into larger decision support frameworks, enabling continuous adaptation without requiring full retraining. This modularity makes TigCLaF highly versatile for dynamic environments. Furthermore, edge deployment of LoRA and QLoRA fine-tuned LLaMA models has demonstrated that sentiment classification can run efficiently on low-power hardware, making advanced NLP accessible in resource-limited environments where computational resources are scarce, thus democratizing sentiment analysis capabilities across regions with minimal infrastructure.

4.4 Limitations and Ethical Considerations

Despite the promising results of the TigCLaF framework, several limitations remain. First, the model's performance is influenced by the limited size and domain coverage of the available Tigrigna corpus, which may constrain its generalization across dialects, informal text, or emerging slang. Second, the cross-lingual transfer from English and Amharic may introduce semantic drift or bias due to linguistic and cultural differences between source and target languages. Additionally, the reliance on pretrained large language models raises computational and accessibility barriers for institutions with limited resources. From an ethical standpoint, the dataset was carefully curated to avoid personally identifiable information and offensive content. However, bias propagation from upstream models or sentiment labels may still affect fairness across demographic groups. Therefore, responsible deployment requires continuous monitoring, bias auditing, and transparent model reporting. Future work should emphasize community-driven data collection, open benchmarking, and equitable resource sharing to promote inclusive NLP research for underrepresented languages.

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5 Conclusion and Future Work

In this work, we introduced TigCLaF, a novel cross-lingual large language model framework designed to enable sentiment-aware text classification in the low-resource Tigrigna language. The framework bridges linguistic and resource gaps through a unified adaptation pipeline that integrates tokenizer extension, cross-lingual continual pretraining, and parameter-efficient fine-tuning, forming a cohesive adaptation pipeline that bridges linguistic and resource disparities between Tigrigna and high-resource languages. TigCLaF enhances multilingual models by expanding their vocabulary coverage for Ge'ez-based tokens, thereby improving the representation of Tigrigna morphology and orthography. Through masked language modeling and cross-lingual contrastive learning, the model effectively transfers semantic knowledge from high-resource languages such as English and Amharic. Furthermore, the LoRA-based parameter-efficient tuning strategy enables the model to specialize for sentiment analysis with minimal computational overhead, making it highly suitable for low-compute and data-scarce environments. The experimental results demonstrate that TigCLaF significantly outperforms state-of-the-art multilingual models such as XLM-RoBERTa, AfriBERTa, and LLaMA-2 across multiple evaluation settings. The model achieved consistent gains in accuracy and Macro-F1, particularly on morphologically complex and code-switched samples, with substantially reduced computational and parameter costs. Notably, TigCLaF maintained strong performance in few-shot and zero-shot scenarios, highlighting its adaptability and efficiency in constrained environments. The framework advances the broader goal of inclusive AI by ensuring that languages with limited digital presence can benefit from modern NLP technologies while minimizing computational and data demands. Beyond Tigrigna, the TigCLaF framework establishes a reproducible and scalable paradigm for adapting large language models to other underrepresented African and low-resource languages, providing a foundation for future research in cross-lingual transfer learning and culturally grounded natural language understanding. However, limitations remain in handling the complexity of code-switching and orthographic variations, suggesting opportunities for improved multilingual tokenizers and adaptive pretraining strategies.

This study suggested that future research should focus on exploring real-time sentiment analysis applications for social media monitoring and adaptive integration into decision-support systems, and on several key directions to further enhance the TigCLaF framework. Firstly, we plan to extend the model to include more low-resource African languages and evaluate its adaptability across diverse linguistic typologies and dialects. This will involve collecting richer, more varied corpora for languages such as Oromo, Amharic, and Twi, as well as more robust sentiment lexicons. Secondly, we aim to explore multimodal sentiment analysis, integrating image and text data for sentiment-aware classification in social media contexts. This could be achieved by combining vision-based models with TigCLaF, enabling a broader scope of real-world applications. Thirdly, model explainability and bias mitigation will be a priority, as we seek to develop methods for more transparent decision-making processes that address concerns about fairness in sentiment prediction across different social and

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demographic groups. Additionally, fine-tuning strategies will be optimized for resource-constrained devices, expanding the reach of TigCLaF to edge devices with limited computational power. Finally, we plan to contribute to the development of large-scale multilingual benchmarks, integrating Tigrigna and similar languages into existing evaluation frameworks to push forward the research on low-resource NLP in African and underrepresented languages.

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Declarations

Author Contributions

Hagos Gebremedhin Gebremeskel primarily led the study, including background research, model development, analysis, and manuscript writing. Chong Feng supervised the research, contributed to the conceptual design, and provided critical feedback to ensure methodological rigor and clarity. Geleta Negasa Binegde evaluated the TigCLaF framework on Afan Oromo and Somali to assess cross-lingual generalizability. Assefa Mebrahtu contributed to manuscript editing, proofreading, and improvements to presentation. All authors reviewed and approved the final manuscript.

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Competing Interests

The authors declare no competing interests.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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